

RODERICK LANDRETH

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TECHNICAL SKILLS

- **CAD & Design:** SolidWorks, GD&T, Simulink, Microcontrollers, KiCAD PCB design, Jama Connect, 3D printing and Rapid Prototyping with sheet metal, Nylon, PLA, and PETG
- **OS and Software:** Windows, Ubuntu, CentOS 8, VMware, Docker, Jira, Microsoft 365 Suite, L^AT_EX
- **Programming and Scripting:** Python, MATLAB, Bash Scripting, Java

WORK EXPERIENCE

Mechanical Engineer II

at Sherpa 6 Inc.

JUN 2020 - Present

Natick, MA and San Francisco, CA

- Lead design engineer creating the electrical, heat management and component mounting systems for different carry-on, ruggedized, air-cooled edge-processing and networking enclosures with switching power sources, one of which to MIL Aviation specifications and testing.
- Lead engineer for experimental design, testing, and data analysis comparing multiple force sensors' data output and programmatically verifying proprietary algorithms. Produced a research paper.
- Produced a ruggedized leave-behind tactical radio enclosure, 80+ hours operation time, manufactured 20 used in a tropical rain forest enabling network communications for several events.
- Designed ruggedized modular payload system to integrate networking architecture, batteries, chargers, and UAV onto in-field terrain-worn robotics. Fabricated electronics with KiCad.
- Managed and developed a project's requirement ecosystem in Jama Connect w/ Jira integration.
- Travel 25% of the year on integration team troubleshooting network and hardware at integration events.
 - Summit Award (Company's Top Honor) recognized for field support. (Project Convergence 21)
 - Spot Award for field integration and support. (May 2026)

Engineering Intern

at Greno Industries

JUN - AUG 2019

Scotia, NY

- Modeled parts contracted for fabrication with SolidWorks, produced drawings with callouts and GD&T.
- Optimized the procedures used by manufacturing and quality control to validate fabricated parts.

IT Technician Intern

at TwinHats

JUL - SEP 2018

Hopkinton, MA

- Installed domain controllers and active directory, updated 2008/12/16 Windows server architecture.
- Troubleshoot client IT problems and developed assembly-line streamlining software with visual C#.

RELEVANT PROJECTS

Personal Projects

- Modeling and fabrication of primarily 3D printed modular Twin-engine differential thrust RC plane.
- Produced a script to dimension a plane from desired wingspan, wing/tail configurations, and payload.

Union College SAE Aero Competition Team

2016-2020

- Finished third place overall (first among domestic teams) in the international SAE AERO competition.
- Designed a plane able to assemble from a fixed volume in under 3 minutes maximizing payload.
- Senior Project Design Lead
 - 10ft wingspan plane with static and dropped payloads releasing autonomous gliders. Designed, fabricated, and calculated static and dynamic stability. Managed a team budget of \$30,000.

EDUCATION AND CERTIFICATIONS

- **Union College**, Schenectady, NY. B.S. Mechanical Engineering (Honors), Physics Minor. 2016 - 2020
- **CSWP Certified**, Eagle Scout